

Statistical Efficiency of Birth-Death Tree Parameter Estimation

Stat 700

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The Phylodeep Package

Phylodeep Paper is by J. Voznica et al [1]

- ▶ Parameter estimation is difficult and MLE method can be challenging for larger trees.
- ▶ Major components of the paper to address this
 - ▶ CBLV representation of trees
 - ▶ Pre-trained neural networks for pre-specified tree types
- ▶ The goal is to speed up inference

Usage

- ▶ Input: Phylogenetic tree in Newick format
- ▶ Output: Estimated parameters (e.g., birth rate (λ), death rate (μ), R_0)
- ▶ Given a tree, Phylodeep selects the pre-trained model with the highest probability
- ▶ Outputs actually come with uncertainty estimates from a parametric bootstrap

Tree Generation

- ▶ Trees were generated in parallel using the 'treesimulator' package
- ▶ Explored Phylopomp for this, but format is slightly different and packages weren't quickly interoperable

```
1 # Generate a BD tree with 200-500 tips,  
2 # transmission rate of 0.5, removal rate  
3 # of 0.25 and sampling probability of 0.3  
4 generate_bd --min_tips 200 --max_tips 500 \  
5   --la 0.5 --psi 0.25 --p 0.3 \  
6   --nwk tree.nwk --log params.csv
```

Maximum Likelihood Estimation: Overview

- ▶ **Goal:** Estimate birth rate (λ) and death rate (μ) from phylogenetic trees
- ▶ **Method:** Numerical optimization of Stadler (2010) likelihood
- ▶ **Key Feature:** Works with **tree data only** - no nucleotide sequences required
- ▶ **Implementation:** Uses `scipy.optimize.minimize` with L-BFGS-B algorithm

Advantages:

- ▶ Statistically principled (maximum likelihood)
- ▶ Works on small trees (no minimum tip size requirement)
- ▶ Provides point estimates with optimization diagnostics

Stadler (2010) Likelihood Formulation

For a constant-rate birth-death process with sampling probability ρ :

$$\begin{aligned}\ell(\lambda, \mu | T, \rho) = & (n - 1) \log(\lambda) \\ & - (\lambda + \mu) T_{\text{total}} \\ & + n \log(\rho) \\ & - r \cdot T + \text{penalty terms}\end{aligned}$$

where:

- ▶ n = number of tips
- ▶ T_{total} = total branch length
- ▶ T = tree height (time from root to present)
- ▶ $r = \lambda - \mu$ = net diversification rate
- ▶ ρ = sampling probability

Key insight: Survival probability term $(-r \cdot T)$ helps identify μ separately from λ

MLE Implementation: Tree Statistics Extraction

Extracted from each tree:

- ▶ Number of tips: n
- ▶ Total branch length: $T_{\text{total}} = \sum_{e \in E} \ell(e)$
- ▶ Tree height: $T = \max_{v \in V} d(\text{root}, v)$
- ▶ Branching times: All internal node ages
- ▶ Mean branch length: $\bar{\ell}$

Initial parameter estimates (method-of-moments):

$$\hat{r} = \frac{\log(n)}{T}$$

$$\hat{\lambda} = \frac{n-1}{T_{\text{total}}}$$

$$\hat{\mu} = \max(0.01, \hat{\lambda} - \hat{r})$$

These serve as starting values for numerical optimization.

MLE Implementation: Optimization

Numerical optimization:

- ▶ **Algorithm:** L-BFGS-B (bounded optimization)
- ▶ **Bounds:** $\lambda \in [0.01, 10.0]$, $\mu \in [0.01, 10.0]$
- ▶ **Constraint:** $\mu < \lambda$ (enforced via penalty)
- ▶ **Gradient:** Analytical gradient provided for faster convergence

Gradient components:

$$\frac{\partial \ell}{\partial \lambda} = -\frac{n-1}{\lambda} + T_{\text{total}} + T$$

$$\frac{\partial \ell}{\partial \mu} = T_{\text{total}} - T + \text{penalty term}$$

Regularization: Penalty term prevents μ from collapsing to lower bound in large trees

MLE Implementation: Key Features

1. Survival Probability Term

- ▶ Includes $-r \cdot T$ term to help identify μ
- ▶ Accounts for probability of observing n tips given λ and μ
- ▶ Critical for separating μ from λ in the likelihood

2. Regularization for Large Trees

- ▶ Penalty: $-0.1 \log(\mu)$ when $\mu < 0.1$ and $n > 10$
- ▶ Prevents μ from getting stuck at lower bound (0.01)
- ▶ Encourages realistic death rate estimates based on tree size

3. Robust Error Handling

- ▶ Checks for invalid trees ($n \leq 1$, $T_{\text{total}} \leq 0$)
- ▶ Handles optimization failures gracefully
- ▶ Returns success status, log-likelihood, and iteration count

4. Method-of-Moments Initialization

- ▶ Smart starting values: $\hat{r} = \log(n)/T$, $\hat{\lambda} = (n-1)/T_{\text{total}}$
- ▶ Ensures faster convergence and avoids local minima

MLE: Advantages and Limitations

Advantages:

- ▶ Statistically principled (maximum likelihood)
- ▶ No minimum tree size requirement
- ▶ Works on small trees ($n < 50$)
- ▶ Provides optimization diagnostics
- ▶ Fast computation (seconds per tree)
- ▶ Based on well-established Stadler (2010) theory

Limitations:

- ▶ Higher RMSE than PhyloDeep on large trees
- ▶ Point estimates only (no uncertainty quantification)
- ▶ Requires careful initialization for convergence
- ▶ Can struggle with very large trees ($n > 500$)

Best use case: Small to medium trees where statistical rigor is important

Phylodeep Methodology

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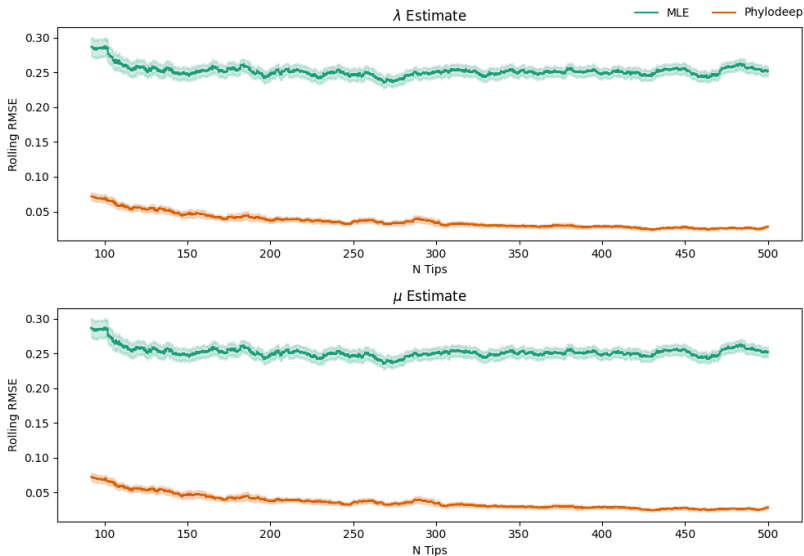


Figure: Rolling bootstrapped RMSE estimates of μ and λ by tree size and resulting 95% CI. Window size is 500.

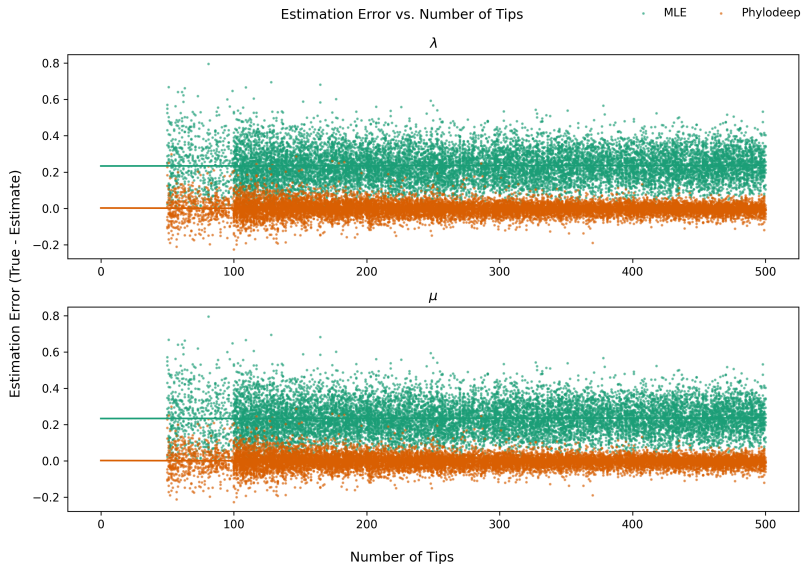


Figure: Raw plot of the absolute errors in λ, μ by tree size.

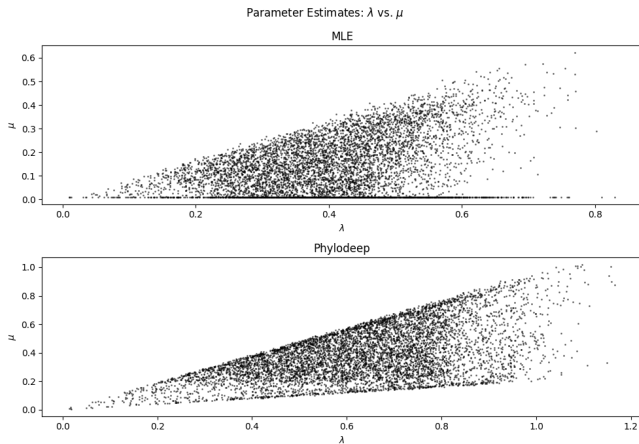


Figure: Plot of λ vs μ estimates to show distribution of estimates.

Main takeaways:

- ▶ RMSE decreases with increasing tree size for both methods.
- ▶ MLE can be used on smaller trees - Phylodeep requires a minimum tip size of around 50.
- ▶ Otherwise, Phylodeep has lower RMSE than MLE, probably due to being trained on 3.9M trees.

References

- [1] J. Voznica et al. “Deep learning from phylogenies to uncover the epidemiological dynamics of outbreaks”. en. In: *Nature Communications* 13.1 (July 2022). Publisher: Nature Publishing Group, p. 3896. ISSN: 2041-1723. DOI: 10.1038/s41467-022-31511-0. URL: <https://www.nature.com/articles/s41467-022-31511-0> (visited on 11/04/2025).